

Robbie Chia Yuan Ee

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EDUCATION

National University of Singapore

B.Sc. in Data Science & Analytics, Minor in Computer Science

- GPA: 4.64/5.00 | NUS Merit Scholarship
- Relevant Coursework: Linear Algebra, Calculus, Probability & Statistics, Data Visualisation, AI & Machine Learning, Data Structures & Algorithms, Database systems, Optimisation Techniques

Singapore

Apr 2022 – Current

TECHNICAL SKILLS

Languages: Python, SQL (PostgreSQL), R, familiar with Java and C

Data Science, ML & AI: Scikit-learn, TensorFlow, JAX, PyTorch, Bayesian Optimisation, NLP, Time-series

LLM & GenAI: Familiar with LLMs, Transformers, RAG, MCPs

Analytics & Visualisation: Pandas, Numpy, SciPy, Matplotlib, Seaborn, Plotly, Streamlit, ggplot2, Microsoft Power BI

Tools: Git, Microsoft Excel, Docker, NLTK, familiar with Apache Spark

EXPERIENCE

Data Analyst Intern

May 2024 – Dec 2024

Auristone Pte Ltd

Singapore

- Designed and implemented an end-to-end RNA-seq ETL and statistical analysis pipeline in R, standardising preprocessing, normalisation and differential expression workflows across experiments
- Formulated and tested statistical hypotheses on treatment-control gene expression differences using EDA, clustering and distributional analysis to identify biologically significant patterns
- Implemented quality control and concordance validation using correlation metrics and outlier detection to detect data inconsistencies and prevent propagation of experimental artefacts into downstream modelling
- Engineered scalable data extraction workflows using MinIO object storage to support repeated modelling experiments on shared high-volume datasets

Operations & Analytics Officer

Apr 2021 – May 2022

Republic of Singapore Air Force (RSAF)

Singapore

- Designed Power BI dashboards tracking operational KPIs to reduce reporting cycles by 60%, enabling stakeholder management to monitor trends, identify anomalies or compliance gaps and support data-informed decision-making
- Digitalised manpower tracking using NFC systems and AWS S3, removing manual logging and improving data reliability

PROJECTS

Uni Curriculum-Workforce Alignment Tool | *Python, SQLAlchemy, sentence-transformers, Plotly, NLTK*

- Built an end-to-end NLP pipeline measuring Singapore workforce preparedness (22k postings) from university curriculum (7000 modules), combining BM25, sentence-transformer embeddings and reciprocal rank fusion to produce signals for curriculum policy review
- Designed a hybrid module-job relevance scoring system with thresholds calibrated against 4000 manually annotated samples, producing interpretable degree-role fit scores for non-technical policy stakeholders
- Delivered a Streamlit dashboard and structured report exports enabling policy officers to explore skill gaps, degree curriculum pitfalls, and role-specific module recommendations across 17 job role families

Character-level LLM | *Python, JAX, Optuna, Flax, Pickle*

- Built a small character-level Transformer from scratch in JAX/Flax with multi-head self-attention, sinusoidal/rotary/ALiBi positional encodings, and feed-forward blocks, achieving $\approx 70\%$ test accuracy
- Designed a four-phase experimental pipeline with fixed-step budgets, token throughput tracking (tokens/sec)
- Conducted single-variable ablation studies over 10+ architectural choices and applied Bayesian hyperparameter search (Optuna/TPE) to identify optimal learning rate and weight decay regions

Fraud Detection in Electricity & Gas Usage | *Python, scikit-learn, LightGBM, Optuna*

- Designed per-fold SMOTE resampling within cross-validation splits to prevent minority-class leakage into validation sets; used fraud recall as the primary metric on a severely class-imbalanced dataset (5.58% fraud rate)
- Compared six classifier families with Optuna-tuned hyperparameters; Bidirectional LSTM on invoice consumption history achieved Fraud Recall of 0.67, outperforming Logistic Regression (best linear baseline) by a clear margin

ADDITIONAL EXPERIENCE

Residential Assistant

Jul 2024 – Present

Residential College 4, National University of Singapore

- One of 15 primary points of contact responsible for student welfare and operational support for a 600-resident hall
- Coordinated with resident faculty, campus security, and housing services to enforce housing policies and manage emergency incident responses